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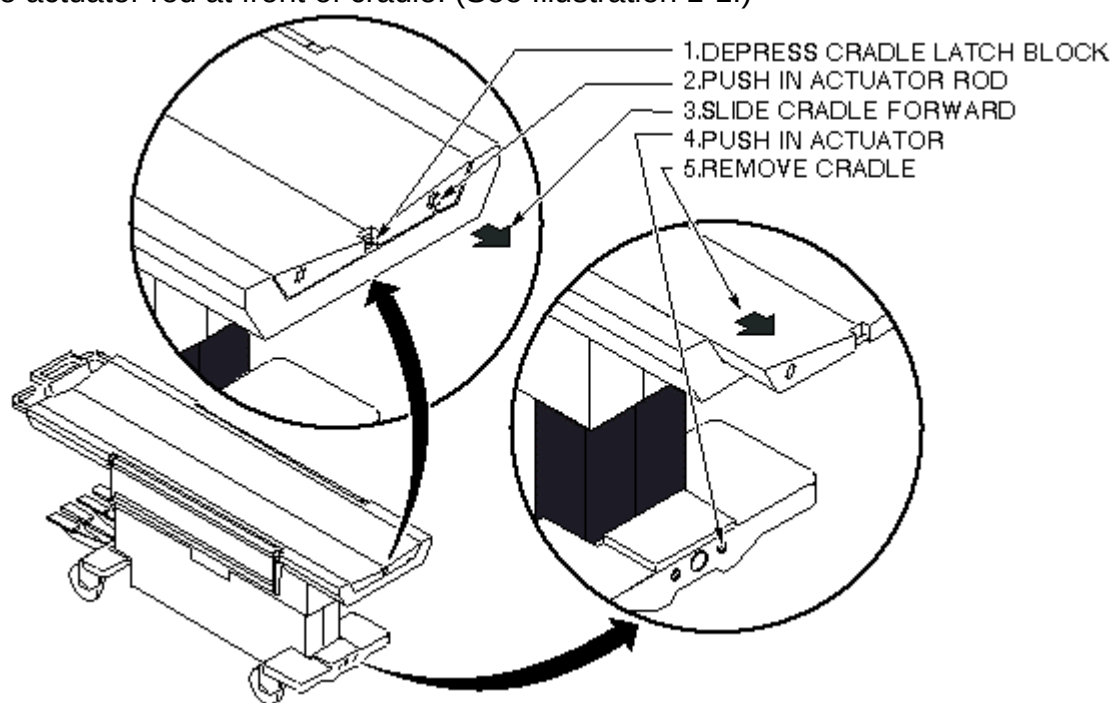
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Description - Contains all adjustment procedures for the Lightweight Patient Transport.

1- REMOVING CRADLE ASSEMBLY

This Section contains all adjustment procedures that may need to be performed on a Lightweight Patient Transport. However, most of the adjustments are factory preset and should not require adjusting in the field. The adjustments should only be performed: 1) as part of the installation specified by the installation procedures, 2) after determination of a problem via the functional check procedures in this manual, or 3) after a component replacement as specified in the replacement procedures in this manual.

1. At the same time, depress the cradle latch block at the front end of transport and push in the actuator rod at front of cradle. (See Illustration 1-1.)



PATIENT TRANSPORT CRADLE REMOVAL
ILLUSTRATION 1-1

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2. While holding these actuators, slide cradle forward on patient transport approximately three inches. The actuators can now be released.
3. Push in the right actuator in the base at front end of transport, and slide cradle the rest of the way off the patient transport. (See Illustration 1-1.)

2- LEVELING PATIENT TRANSPORT

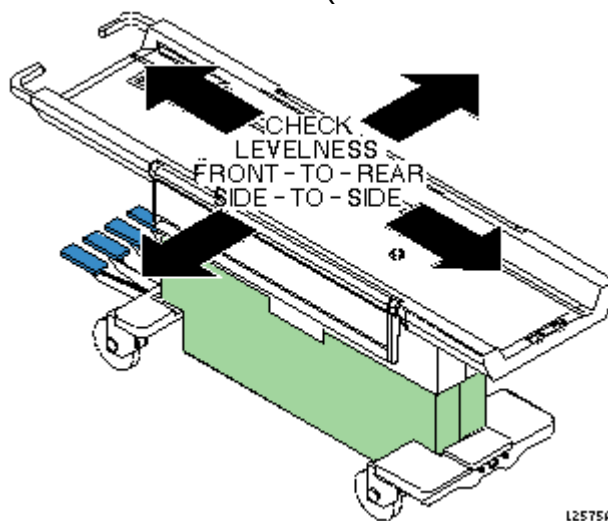
2-1 Required Tools

- 46-194427P255: Torque Wrench, Ratchet Type, 3/8 inch drive, 30-200 in-lb
- 46-307811P1: Socket Driver, Hex Allen-Head, 3/8 inch drive, 5/32 inch bit

2-2 Checking Levelness of Patient Transport

The Patient Transport is leveled by adjusting caster height. However, caster height must also be checked for uniform loading.

1. Dock the Patient Transport. If docking mechanism is not already properly adjusted, position the Patient Transport where it will be located when it is docked.
2. Lock the caster brakes.
3. With table top in fully raised position, check Patient Transport for levelness, both front-to-back and left-to-right. This should be checked at the table top level, not at the table base, since table top alignment is the critical factor. (See Illustration 2-1.)



CHECKING LEVELNESS OF PATIENT TRANSPORT
ILLUSTRATION 2-1

4. Check that all casters are sharing the load approximately equally by applying a small lifting force at each corner of the table and seeing if the caster easily lifts off the floor.
5. If one caster or two diagonally mounted casters seem to not be loaded as much as the other casters, they will require adjustment.

2-3 Checking Caster Height Adjustment

1. To help keep track of measurement data, temporarily label each caster as 1, 2, 3 and 4. Suggestion: label steering caster (green-colored locking tab) as 1, and proceed labeling in a clockwise direction.
2. At each of the four caster wheel assemblies, measure the height from floor to top of aluminum support casting (measurement "A" in Illustration 2-2).
 - a. Record this measurement in Table 1, MEASUREMENT A column labeled 1ST HEIGHT. This height should be 11.0 inches, ± 0.25 inches (279.4 mm, ± 6.4 mm).
 - b. If the 1st Height measurement is **not** within tolerance, calculate amount by which it is out of tolerance (+ if caster height is too small; - if caster height is too large). Record this value in 1ST DELTA column under MEASUREMENT A.

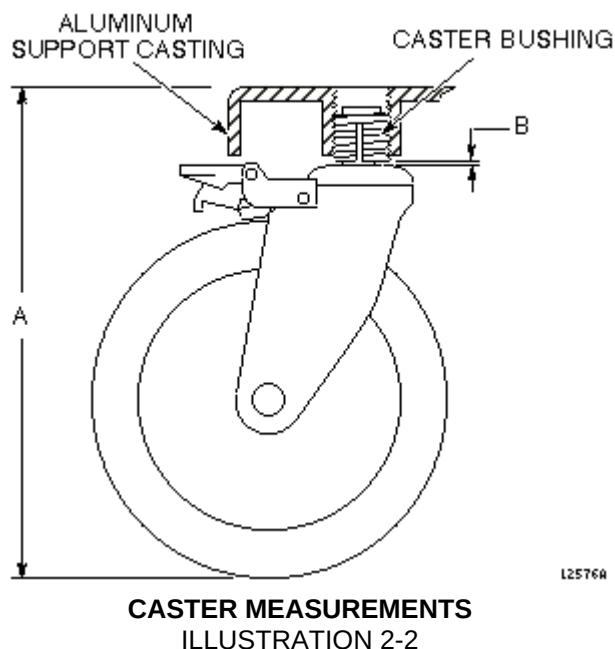


TABLE 1
MEASUREMENT A

MEASUREMENT B

CASTER	HEIGHT	1ST		2ND		GAP	DELTA $\pm$
		DELTA $\pm$	HEIGHT	DELTA $\pm$			
1							
2							
3							
4							

Note:

Tolerance for Measurement A: 11.00, ± 0.25 inch

Tolerance for Measurement B: < 0.25 inch

3. At each of the four caster wheel assemblies, measure the gap between bottom of caster bushing and the base of caster shaft (measurement "B" in Illustration 2-2).

- a. Record this measurement in Table 1, MEASUREMENT B column labeled GAP. The acceptable measurement for this gap is less than 0.250 inches (6.4 mm).
- b. If Gap is **not** within tolerance, calculate amount by which it is out of tolerance (+ if gap is too small; - if gap is too large). Record this value in Delta column under Measurement B.

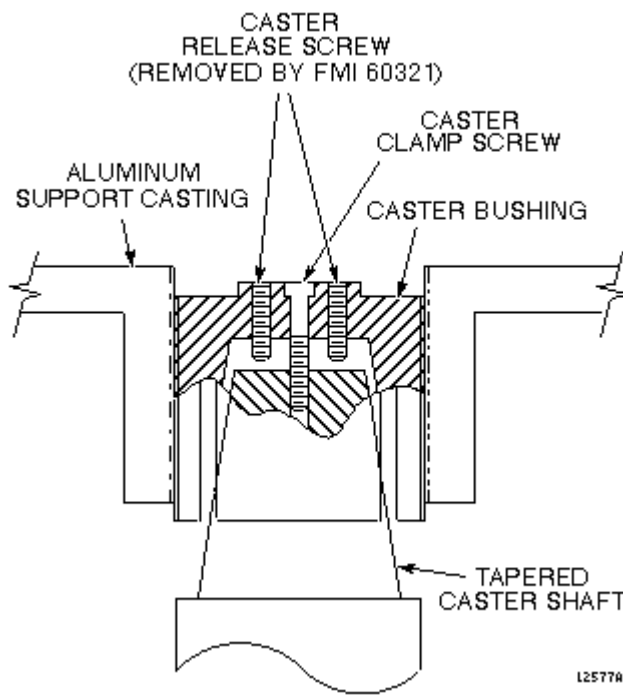
2-4 Caster Mounting Hardware Design Description

When the caster bushing can turn freely inside the aluminum support casting, the bushing provides a means of adjusting caster height. The bushing is slotted and has a tapered hole to accept the tapered caster shaft. (See Illustration 2-3.)

When the caster clamp screw is inserted through the bushing and tightened securely into the caster shaft, the shaft is drawn up into the bushing causing the bushing to spread tightly against the aluminum support casting threads, and preventing the bushing from turning and changing the height adjustment.

To adjust caster height:

- The clamp screw is backed out of the caster shaft and the caster release screws (or 2 caster clamp screws after FMI 60321 is performed) are turned into the caster bushing. The release screws push the caster shaft out of its tapered fitting in the caster bushing, leaving the bushing loose enough to turn and provide caster height adjustment.
- After adjustment, the release screws must be backed out to prevent contact with the caster shaft. The clamp screw can then be adequately tightened, securing the caster shaft and immobilizing the bushing.



CASTER MOUNTING CONSTRUCTION
ILLUSTRATION 2-3

There are two conditions that could cause the caster assembly to come loose and possibly fall off the Patient Transport. These two conditions, their causes, and solutions are described below.

Conditions and Causes:

- Tapered caster shaft can become loose and back out of the caster bushing. This condition can occur if the caster clamp screw was not adequately tightened or is missing altogether. This condition can also be caused if the caster release screws were left in too far before tightening the caster clamp screw, while adjusting caster height. The screws would be pushing against the caster shaft, preventing the shaft from seating securely into the bushing.
- Caster bushing can back out (partially or completely) of its threaded hole in the support casting. If the tapered caster shaft becomes loose (as described above), or if the caster bushing was improperly positioned during caster height adjustment, the bushing can back out of the aluminum support casting. (Correct bushing positioning is detailed in *Section 2-5 Patient Transport Leveling*.)

Solutions:

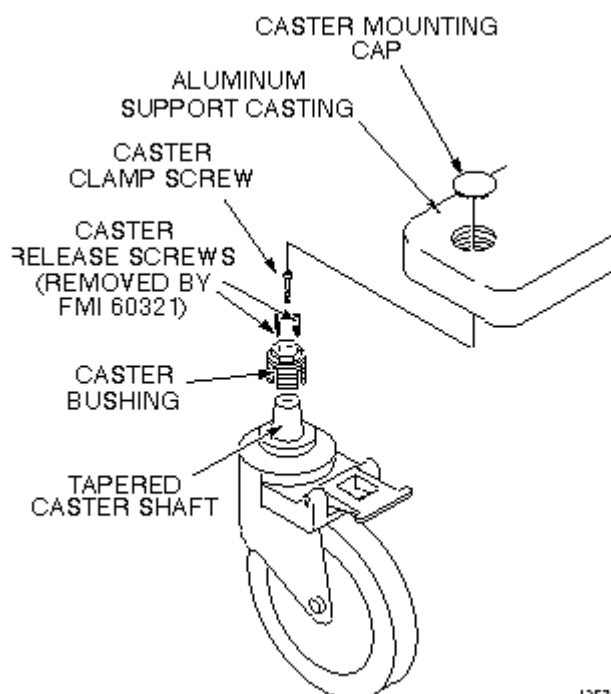
- To avoid the condition caused by leaving the caster release screws in too far, FMI 60321 eliminates the caster release screws from the caster assembly hardware. (Currently being removed from all forward production Patient Transports.) When it becomes necessary to force the tapered caster shaft out of the caster bushing during an adjustment procedure, two caster clamp screws will be used in the vacant caster release screw holes. One of the caster clamp screws will come from the caster being adjusted, and one will be borrowed from another caster assembly.
- 50 in-lb torque requirement for the caster clamp screw. This torque on the caster clamp screw will result in sufficient tightness of the caster bushing inside the aluminum support casting to prevent the bushing from backing out when the caster assembly swivels. This torque requirement is currently in effect for all forward production Patient Transports and is required for FMI 60321.

2-5 Patient Transport Leveling



FERROUS MATERIAL HAZARD! TORQUE WRENCH AND SOCKET DRIVE CONTAIN FERROUS MATERIAL AND CAN BECOME FORCIBLY ATTRACTED TO THE MAGNET. DO NOT BRING THESE TOOLS INTO THE MAGNET ROOM.

1. The torque wrench and socket driver **cannot** be brought into the Magnet Room. Undock the Patient Transport and move it out of the Magnet Room.
2. On first caster to be adjusted, remove the caster mounting cap by prying it off with a small screwdriver. (See Illustration 2-4.)



CASTER WHEEL ASSEMBLY

ILLUSTRATION 2-4

CAUTION

Adjustments that exceed tolerances given in the steps below leave caster bushing vulnerable to damage, which can result in injury to patient and/or persons near the Patient Transport.

3. For each caster **within** tolerance, as determined in Table 1, proceed to Step 14 of this section. For each caster **out of** tolerance (i.e., has 1ST DELTA entry in Table 1), adjust caster height by continuing to the next Step 4.

Note

This procedure assumes FMI 60321, Signa Lightweight Patient Transport Caster Height Adjustment was performed which removes the caster release screws from the caster assembly hardware. If FMI 60321 was not performed, the caster release screws will still be in the screw holes.

4. On the first caster to be adjusted, remove caster clamp screw (cap screw) from its hole, and thread the screw into one of the empty caster release screw holes. Turn caster clamp screw in until finger-tight only.
5. Go to another caster assembly and remove the caster clamp screw. Bring caster clamp screw back to caster being adjusted.
6. At the caster being adjusted, turn the second caster clamp screw into the other empty caster release screw hole until finger-tight only.

7. Alternately screw in caster clamp screws one turn at a time until caster shaft is as loose as possible in caster bushing.
8. Using a means of leverage, slightly prop up the aluminum support casting at the caster being adjusted **only enough** to take Patient Transport weight off caster wheel. (This will disengage caster shaft from caster bushing, and prevent excessive wear on threads during adjustment.)
9. Remove both caster clamp screws from the two caster release screw holes.
10. Adjust caster height an amount equal to the value recorded in 1ST DELTA column of MEASUREMENT A or MEASUREMENT B, whichever pertains, in Table 1.
 - While holding caster wheel, use a 19 mm socket to turn caster bushing counterclockwise to raise table corner (for + DELTA) or clockwise to lower table corner (for - DELTA).
 - Each complete revolution of caster bushing raises or lowers caster 0.083 inches (2.1 mm).
11. Remove prop from under the aluminum support casting.
12. Return borrowed caster clamp screw to its original caster assembly.
13. At caster being adjusted, insert remaining caster clamp screw into caster clamp screw hole.
14. Using the Torque Wrench (46-194427P255) and Socket Driver (46-307811P1), torque caster clamp screw to **50 in-lb**. If necessary, refer to operating instructions supplied with the torque wrench.
15. Verify that caster shaft is seated well into caster bushing and Gap B is still within tolerance defined in Table 1.
16. Move Patient Transport back into magnet room, and dock to the magnet.
17. For each caster, remeasure Height A and record its value in Table 1 in MEASUREMENT A column labeled 2ND HEIGHT. If this 2nd Height value is **out of** tolerance, calculate the amount by which it is out of tolerance, and record this amount in 2ND DELTA column under MEASUREMENT A.
18. For each caster height still out of tolerance, adjust height as determined by value of 2ND DELTA in Table 1 by performing the procedure beginning with Step 4 of this section.
19. When all caster height measurements are within tolerance, replace all caster mounting caps.

3- DOCK HOOK TENSION ADJUST

1. Attempt to dock the Patient Transport, paying attention to the force required to engage the docking mechanism.

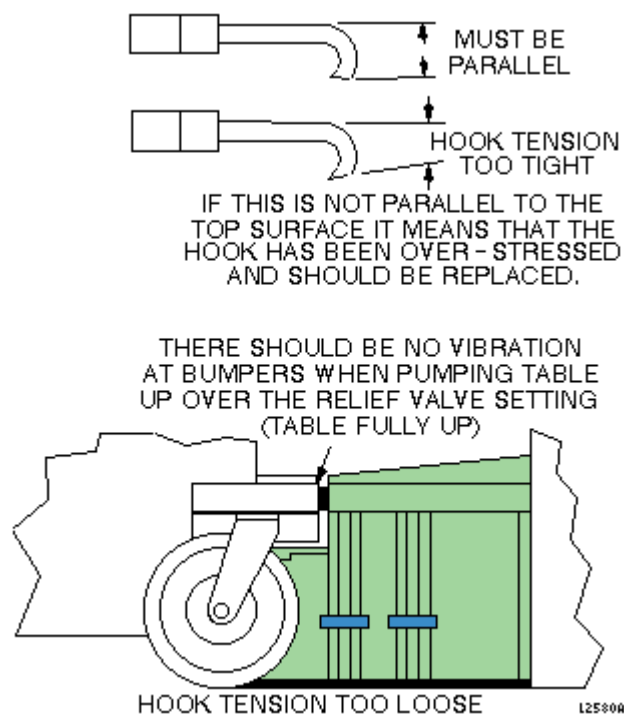
2. If the force to dock seems excessive, stop and loosen the hook (next step). Verify that neither of the problem symptoms indicated in Table 2 is present.

TABLE 2
DOCK HOOK MISADJUSTMENT SYMPTOMS

Symptom	Cause	Action
Excessive force is required to depress dock pedal and fully engage dock latch. (See Illustration 3-1.)	Hook tension too tight.	Loosen hook one turn and Retest.
Excessive back-and-forth vibration of patient transport when using TABLE UP foot pedal at dock; especially noticeable when table top is already in fully raised position. (See Illustration 3-1.)	Hook tension too loose.	Tighten hook one turn and retest.

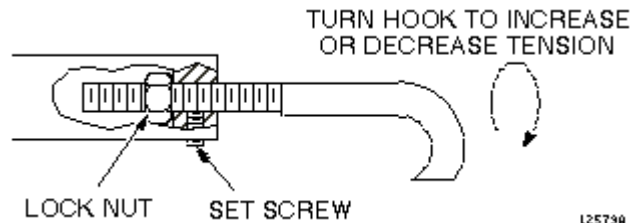
Note

Hook tension adjustment is very sensitive. Never adjust hook more than **one turn at a time** without retesting.



DOCK HOOK TENSION CHECK
ILLUSTRATION 3-1

3. If dock hook tension requires adjusting, loosen set screw and lock nut near base of dock hook, turn hook one turn, tighten set screw and lock nut, and recheck hook adjustment. (See Illustration 3-2.)



DOCK HOOK TENSION ADJUST
ILLUSTRATION 3-2

CAUTION

Do not adjust the dock hook tension more than one turn at a time. Adjusting more than one turn at a time may cause damage to internal Patient Transport mechanisms.

4- DOCK CENTERING

Dock, Hydraulic/Longitudinal Position - For proper Patient Transport docking, the dock assembly must be adjusted to meet two criteria:

1. Gap between patient transport top and front of Magnet Enclosure must be $\frac{3}{4}$, $\pm\frac{1}{4}$ inch (19.0, ± 6.4 mm). This is to ensure:
 - Gap is wide enough so there is no finger pinch hazard when docking transport.
 - Gap is narrow enough so UP LIMIT sensor operates properly.
2. Dock assembly **must** be square to front of magnet, so axis of cradle travel is the same for both Patient Transport and Bridge.

4-1 Longitudinal Adjustment (In and Out)

1. Dock, Hydraulic/Longitudinal Adjustment - Measure distance from patient transport top to front cover. (See Illustration 4-1.) Dimensions A and B should be $\frac{3}{4}$, $\pm \frac{1}{4}$ inch (19.0, ± 6.4 mm) and equal to each other so the patient transport top is colinear with the bridge.

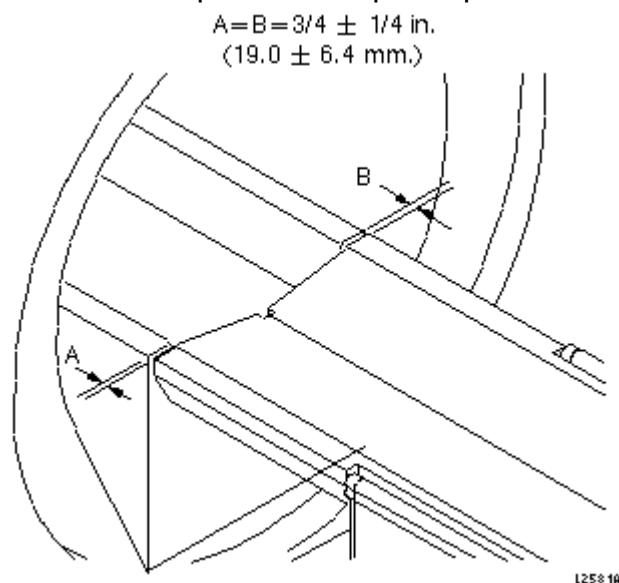
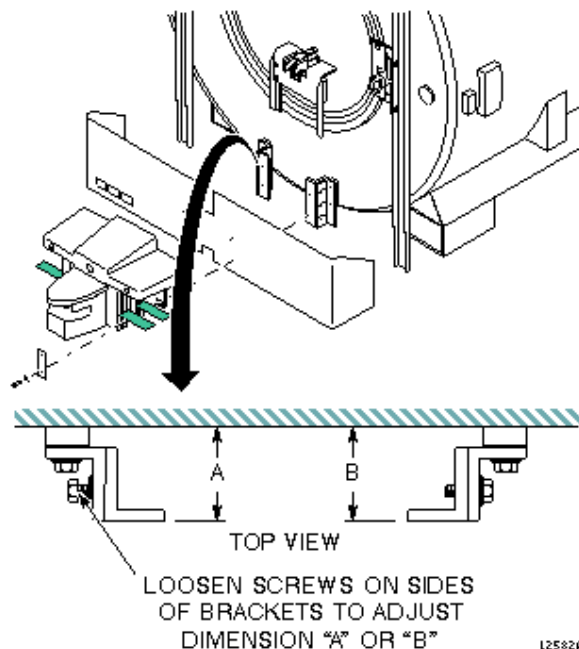


TABLE DOCKING MEASUREMENT FOR DOCK
ILLUSTRATION 4-1

2. If adjustment is required, loosen screws on dock support brackets, move dock to proper position, and tighten screws. (See Illustration 4-2 for location of dock support brackets.)

WARNING!

Dock contains highly ferrous material. Loosen, but do NOT remove, mounting bolts.



DOCK POSITIONING ADJUSTMENT
ILLUSTRATION 4-2

4-2 Lateral Adjustment (Left to Right)

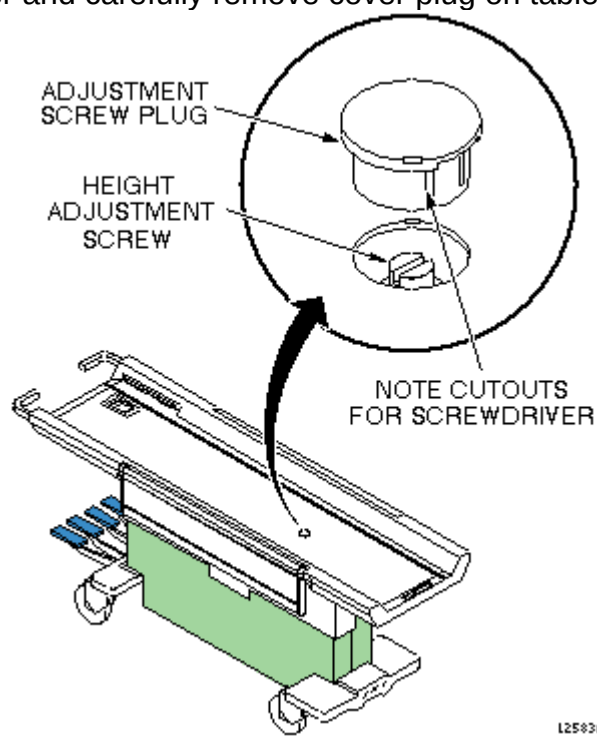
1. Raise table to maximum height using the UP foot pedal.
2. Dock Patient Transport.
3. Check alignment of Patient Transport against bridge. Guide ridge down the length of table top should match guide ridge down length of bridge (left-to-right only; height has not yet been adjusted).
4. If adjustment is required, loosen screws on dock support brackets, move dock to proper position, and tighten screws. (See Illustration 4-2 for location of dock support brackets.)

WARNING!

Dock contains highly ferrous material. Loosen, but do NOT remove, mounting bolts.

5- TABLE TOP HEIGHT ADJUSTMENT

1. Dock Patient Transport.
2. Pump table top to fully raised position.
3. Use a small screwdriver and carefully remove cover plug on table top. (See Illustration 5-1.)



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ILLUSTRATION 5-1

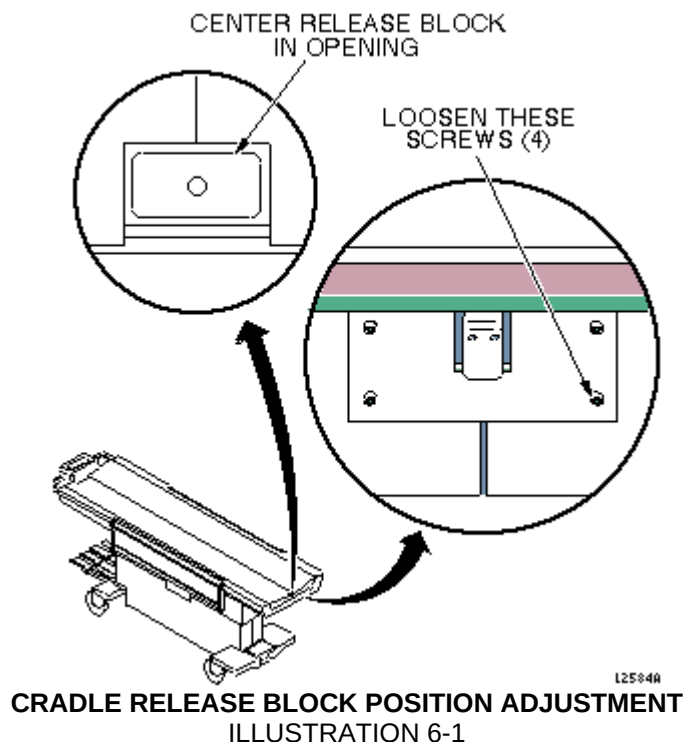
4. Adjust screw in table top to set table top height equal to bridge height.

5. Replace cover plug.

6- CRADLE RELEASE BLOCK ADJUSTMENT

6-1 Cradle Release Block Position Adjustment

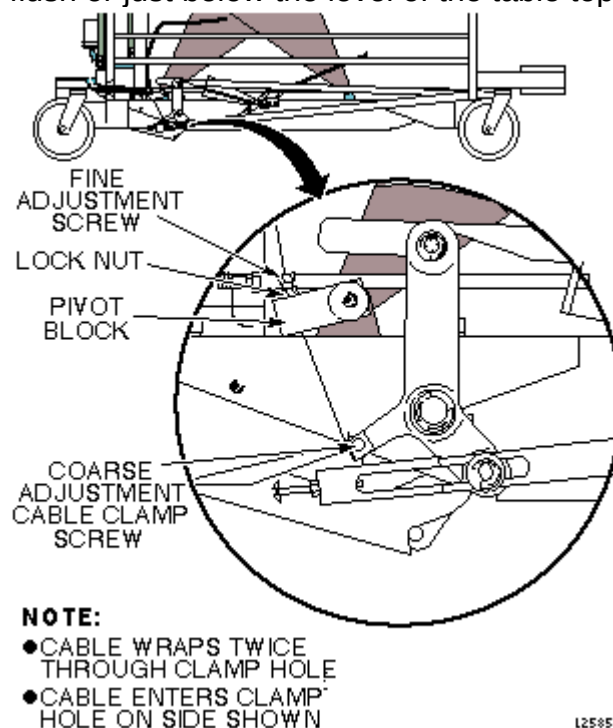
1. Loosen four hex socket-head screws securing the release block assembly. (See Illustration 6-1.)



2. Adjust the position of the assembly so the release block is centered in the cutout on the front of the cradle assembly, and tighten hex socket screws. (See Illustration 6-1.)

6-2 Cradle Release Block Retraction Adjustment

1. Dock Patient Transport.
2. Remove all upper and lower trim covers from sides of Patient Transport.
3. Loosen the lock nut, then tighten or loosen the cable adjustment screw in pivot block so the cradle release block is flush or just below the level of the table top. (See Illustration 6-2.)



CRADLE RELEASE BLOCK RETRACTION ADJUSTMENT
ILLUSTRATION 6-2

4. Tighten lock nut.

Note

If the adjustment screw does not provide enough range for adjustment, coarse adjustment may be made to the cable length where the cable end is clamped into actuator bracket. Fine adjustment will be required after performing a coarse adjustment.

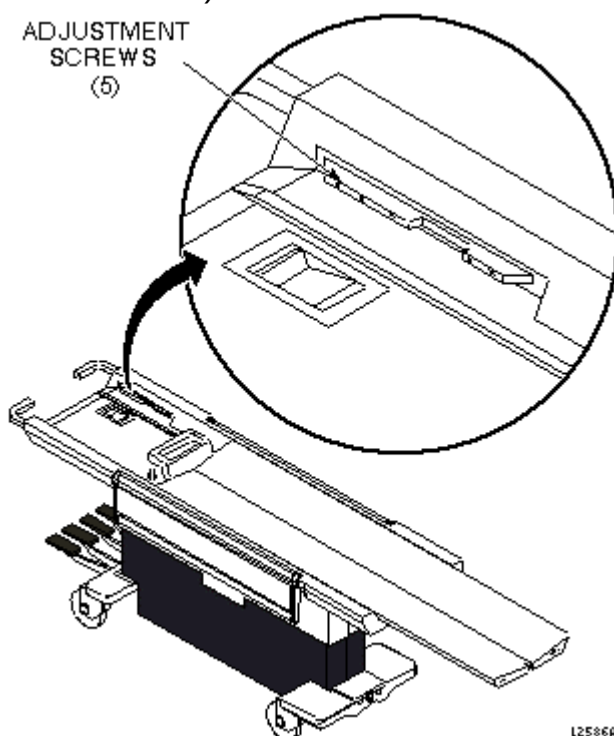
7- CRADLE GUIDE RAIL LATCH ADJUSTMENT

Description

The cradle is locked in home position on the Patient Transport by a latch in the cradle that engages a notch in the cradle guide strip near the end of the Patient Transport. The cradle should be able to just engage this notch, without extra force, when pushed to home position on the table top, yet it must engage it tight enough so the table top will not move on the Patient Transport when it is undocked.

Procedure

1. Refer to *Section 1 - Removing Cradle Assembly* to partially remove cradle from table top.
2. Loosen five screws securing guide rail and slide guide rail forward or backward as required. Tighten screws. (See Illustration 7-1.)

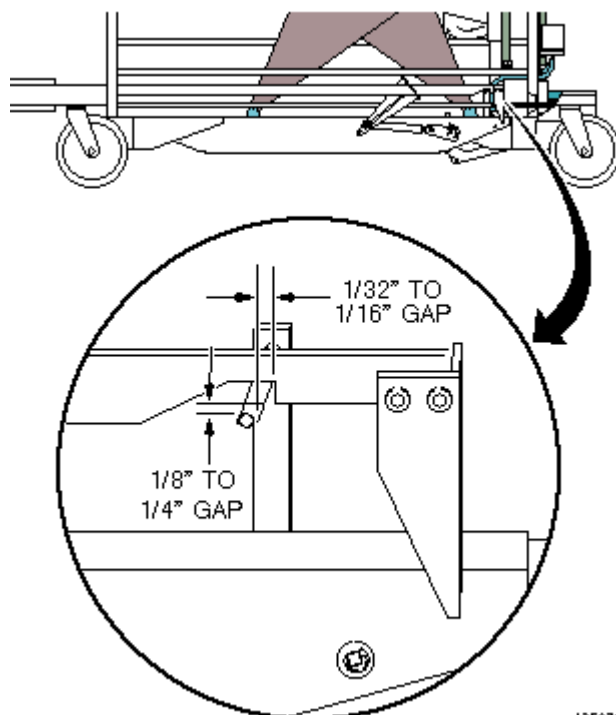


CRADLE GUIDE RAIL LATCH MOD
ILLUSTRATION 7-1

8- CRADLE HOME/TABLE DOWN INTERLOCK ADJUSTMENT

This adjustment is for the mechanical interlock that prevents the Patient Transport from being undocked or lowered while the cradle is in the magnet bore.

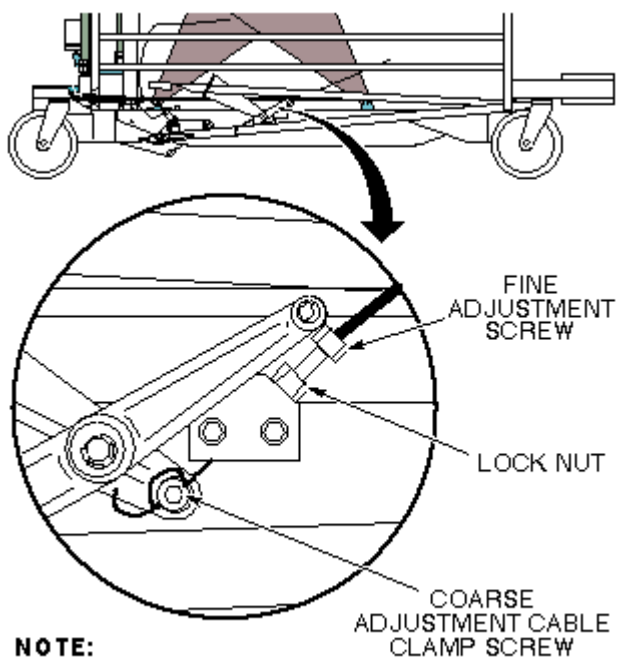
1. Remove all upper and lower trim covers from sides of Patient Transport.
2. With cradle **not** in HOME position (removed), check the interlock mechanisms for tolerances shown in Illustration 8-1. To adjust:
 - a. 1/16 to 1/32 inch (1.59 to 0.79 mm) Gap - Carefully bend interlock rod. (See Illustration 8-1.)



CRADLE HOME/TABLE DOWN INTERLOCK
ILLUSTRATION 8-1

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- b. 1/8 inch (3.18 mm) Gap - Loosen lock nut and tighten or loosen the cable adjustment screw to achieve the proper gap, and tighten lock nut. (See Illustration 8-2.)



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CRADLE HOME/TABLE DOWN INNERLOCK

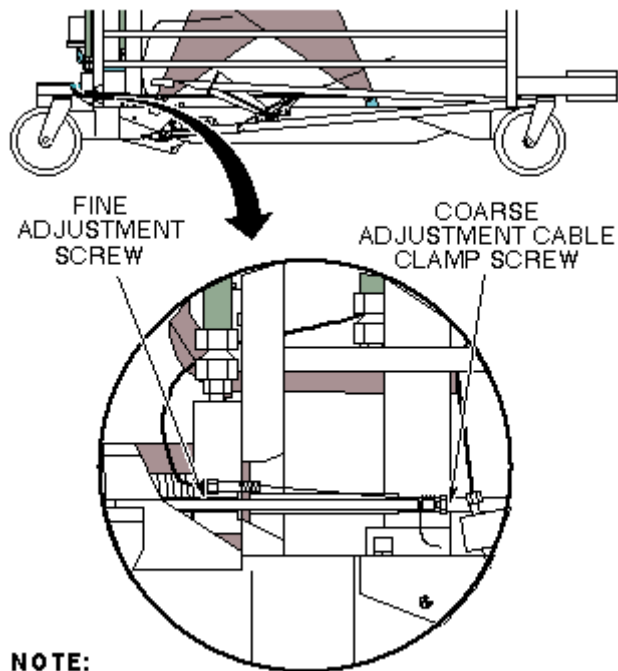
ILLUSTRATION 8-2

Note

If the adjustment screw does not provide enough range for adjustment, coarse adjustment may be made to the cable length where the cable end is clamped into actuator bracket. Fine adjustment will be required after performing a coarse adjustment.

9- CRADLE RELEASE PIN HEIGHT ADJUSTMENT

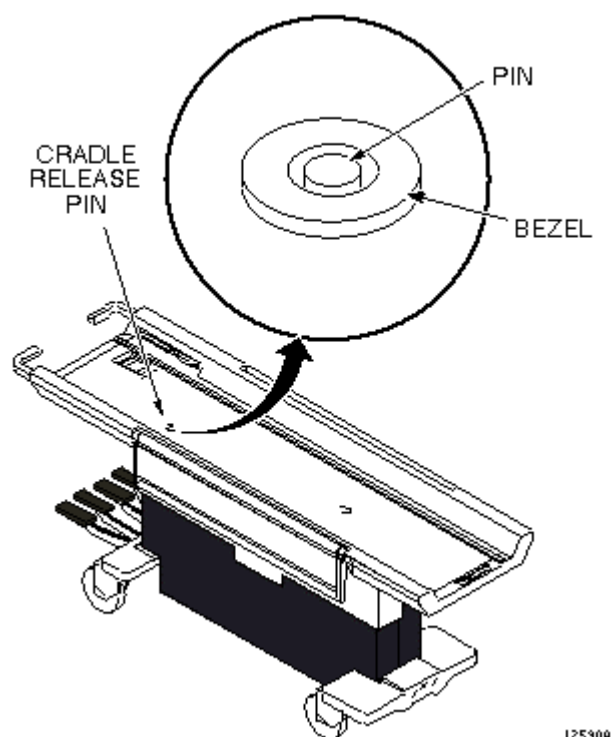
1. Remove Cradle Assembly from Patient Transport as described in *Section 1 - Removing Cradle Assembly*.
2. Remove all outer and inner trim covers from sides of Patient Transport.
3. Tighten or loosen the cable adjustment screw so that the cradle release pin is flush to 1/16 inch (1.59 mm) below the level of the table top. (See Illustrations 9-1 and 9-2.)



NOTE:
CABLE WRAPS TWICE
THROUGH CLAMP HOLE

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CRADLE RELEASE PIN HEIGHT ADJUSTMENT
ILLUSTRATION 9-1



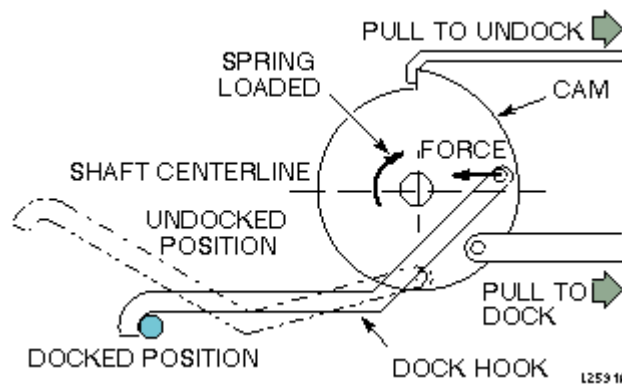
CRADLE RELEASE PIN HEIGHT
ILLUSTRATION 9-2

10- DOCK PEDAL STROKE ADJUSTMENT

Description

The table dock hook operates on the over-center principle. This is implemented by fixing one end of the hook to a cam on a shaft. (See Illustration 10-1.)

- When the force on the hook at its point of attachment to the cam is on the opposite side of the shaft as the hook end, this force will tend to keep the hook in the docked position.
- When the point of force is moved over-center of the shaft centerline, the force will no longer act to keep the hook retracted, and the spring loaded cam will extend the hook to the undocked position.

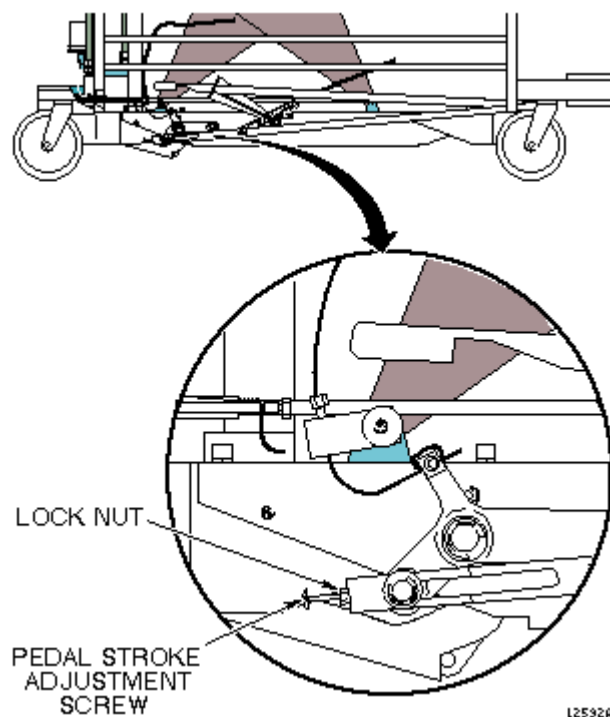


DOCKING MECHANISM DIAGRAM
ILLUSTRATION 10-1

Two style table adjustments are currently in the field. The type of table at the site is determined by manufacturing date. Choose the procedure appropriate for your table style.

10-1 Tables Manufactured Before April 2002

1. Remove all outer and inner trim covers from sides of Patient Transport.
2. Tighten or loosen the stroke adjustment screw as required so when the dock pedal is fully depressed during the docking operation, the hook cam mechanism just goes over-center. (See Illustration 10-2.)



DOCK PEDAL STROKE ADJUSTMENT
ILLUSTRATION 10-2

10-2 Tables Manufactured After April 2002

1. Remove the Hatch at rear of table for adjusting the undock cable. (See Illustration 10-3.)



ILLUSTRATION 10-3

HATCH REMOVAL

2. Loosen the screw counterclockwise, shown in Illustration 10-4, until the table undocks easily when the pedal is pushed. (This will tighten the cable pulling the lever down).

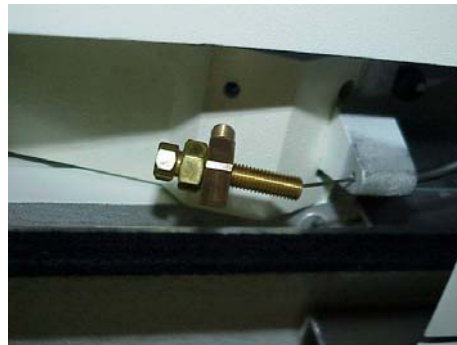
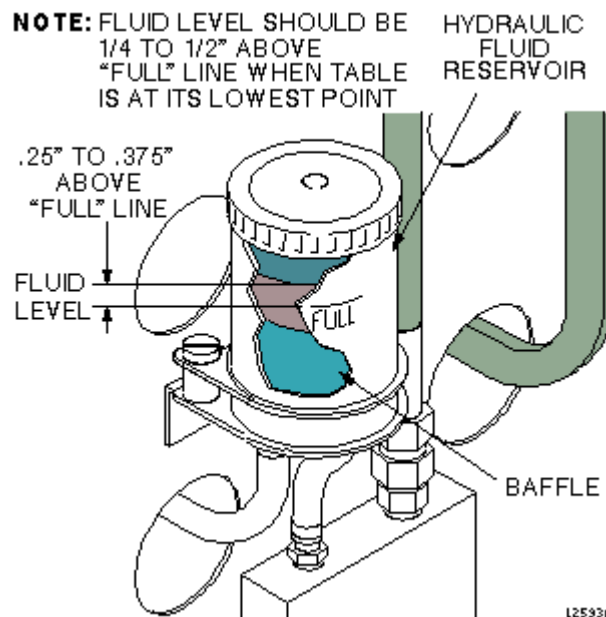


ILLUSTRATION 10-4
CABLE TENSION ADJUSTMENT

3. When finished, tighten the lock nut and reattach the cover. (See Illustration 10-3.)

11- HYDRAULIC FLUID LEVEL CHECK AND FILL

1. Remove all outer and inner trim covers from sides of Patient Transport.
2. Lower table top to FULL DOWN position.
3. Check hydraulic fluid level, and add fluid if necessary. (See Illustration 11-1.)



HYDRAULIC FLUID LEVEL CHECK
ILLUSTRATION 11-1

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	April 17, 1998	W. Pasciak	Initial release converted to Word
0	May 22, 2002	R. Hawthorne	Added new style table dock adjustment in section 10. Updated for rev 0 through out
1	Sept. 30, 2010	R. Kohl	Per CAPA 1859392: Added Warning to Section 4-1 Longitudinal Adjustment and Section 4-2 Lateral Adjustment.